

Validation of the Flaps 2D simulation code

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This report aims to document and verify the “Flaps 2D” simulation tool, developed by myself since February 2012, to investigate the physics of laser-matter interaction in nanostructured materials.

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CONTEXT OF THE SIMULATION TOOL

The development of this work has been supported during 11 months by the french National Research Agency (ANR) under the project “Ultrasonde” leaded by Tatiana Itina, Laboratoire Hubert Curien, Saint-Etienne, France.

Today, I’m still working on the development of this code which, at term, allows to study the absorption by a structured material irradiated by a femtosecond laser. The code has been designed so that material can be also polarized by a high voltage. The coupling of polarized materials with laser illumination allow tomographic reconstruction of atomic lattices in 3D with metals, dielectrics and oxides. The code can also be used to study the transport of free-carriers in nanostructures considering the effect of the geometry. This aspect is also a promiscuous domain for a better understanding of the carrier and heat transport properties at the nanoscale.

CONTENTS

Context of the simulation tool	2
I. System of equations solved by the code	3
II. Free-carrier density: Validation of analytical cases	4
A. Auger recombination	4
B. Interband absorption	4
C. Impact ionization	4
III. Free-carrier energy: validation of analytical cases	4
A. Heating of the interband absorption	4
B. Validation of the electron energy coupling with lattice	5
IV. Validation of lattice thermal coupling	6
A. Heating of the lattice by hot electrons	6
V. Verification of the coupled effects	8
A. Change of carrier heat capacity	8
B. Auger recombination	8
VI. Control of the energy conservation	9
A. Energy of the carriers	10
B. Control of the laser energy	10
1. Calculation of the tip absorptance	10
2. Calculation of the incident laser energy	11
3. Verification of the calculated intensity	12
VII. Generation and control of the mesh	12
VIII. Optimization of the code	13
A. Sequential code: checking the bottlenecks	13
B. Calculation of the code velocity	13
C. Changing the order of the loops	14
D. Parallel code: reduce the number of conditional branches	15
E. Efficiency of the parallelization method	15
References	16

I. SYSTEM OF EQUATIONS SOLVED BY THE CODE

The drift induced by the external field and by the charge separation is neglected.

a. *Free-carrier generation and transport*

$$\begin{cases} \frac{\partial n_e}{\partial t} + \nabla \cdot \mathbf{J}_e &= G_e - R_e \\ \frac{\partial n_h}{\partial t} + \nabla \cdot \mathbf{J}_h &= G_h - R_h. \end{cases} \quad (1)$$

$$G_{e,h} = \sigma_1 I + \sigma_2 I^2 + \delta_{II}^{e,h} n_{e,h}$$

$$R_{e,h} = R_h^A + R_e^A = C_h^{AR} n_h^2 n_e + C_e^{AR} n_e^2 n_h$$

$$\begin{aligned} \mathbf{J}_{e,h} &= -D_{e,h} \nabla n_{e,h} \\ D_{e,h} &= \mu_{e,h} \frac{k_B T_{e,h}}{e} \frac{\mathcal{F}_{1/2}(\eta_{e,h})}{\mathcal{F}_{-1/2}(\eta_{e,h})}, \\ \mu_{e,h} &= \frac{q_{e,h}}{m_{e,h}^* \nu_{e,h}} \frac{\mathcal{F}_0(\eta_{e,h})}{\mathcal{F}_{1/2}(\eta_{e,h})} \end{aligned}$$

b. *Free-carrier energy*

$$\begin{cases} \frac{\partial}{\partial t} (C_e T_e) + \nabla \cdot (\kappa_e \nabla T_e) = Q_e - \gamma_{e-ph} (T_e - T_s) \\ \frac{\partial}{\partial t} (C_h T_h) + \nabla \cdot (\kappa_h \nabla T_h) = Q_h - \gamma_{h-ph} (T_h - T_s) \end{cases} \quad (2)$$

$$\begin{aligned} \frac{\partial (C_{Si} T_{Si})}{\partial t} &= \nabla \cdot (\kappa_{Si} \nabla T_{Si}) + \gamma_{e-ph} (T_e - T_{Si}) + \gamma_{h-ph} (T_h - T_{Si}) \\ C_{e,h} &= \frac{3}{2} k_B n_{e,h} \left\{ \frac{\mathcal{F}_{3/2}(\eta_{e,h})}{\mathcal{F}_{1/2}(\eta_{e,h})} - \eta_{e,h} \left[1 - \frac{\mathcal{F}_{3/2}(\eta_{e,h})}{\mathcal{F}_{1/2}(\eta_{e,h})} \cdot \frac{\mathcal{F}_{-1/2}(\eta_{e,h})}{\mathcal{F}_{1/2}(\eta_{e,h})} \right] \right\} \\ \kappa_{e,h} &= \frac{k_B^2 n_{e,h} \mu_{e,h} T_{e,h}}{e} \left[6 \frac{\mathcal{F}_2(\eta_{e,h})}{\mathcal{F}_0(\eta_{e,h})} - 4 \frac{\mathcal{F}_1(\eta_{e,h})^2}{\mathcal{F}_0(\eta_{e,h})^2} \right] \\ Q_{e,h} &= \frac{\hbar \omega - E_{gap}}{\hbar \omega} \sigma_1 I + \frac{2\hbar \omega - E_{gap}}{2\hbar \omega} \sigma_2 I^2 \\ &\quad - E_{gap} \delta_{II}(T_{e,h}) n_{e,h} + \alpha_{fcr}^{e,h} I + E_{gap} R_{e,h}^A \end{aligned}$$

c. *Lattice energy*

$$\frac{\partial (C_s T_s)}{\partial t} - \nabla \cdot (\kappa_s \nabla T_s) = \gamma_{e-ph} (T_e - T_s) + \gamma_{h-ph} (T_h - T_s) \quad (3)$$

$$\gamma_{e,h-ph} = C_{e,h} \nu_{e,h-ph}. \quad (4)$$

$$\nu_{e,h-ph}^{-1} = \tau_{ph}^0 \left[1 + \left(\frac{n_{e,h}}{n_{th}} \right)^2 \right]$$

$$C_s(T_s) = 10^3 \rho_{Si} \times (a_0 T_s + a_1 T_s^{a_2})$$

$$\kappa_s(T_s) = 10^2 \left(a_0 + \frac{a_1}{1 + e^{-\frac{T_s - T_c + \frac{w_1}{2}}{w_2}}} \right) \left(1 - \frac{1}{1 + e^{-\frac{T_s - T_c - \frac{w_1}{2}}{w_3}}} \right)$$

Since the Fermi-Dirac statistics strongly reduce the possible comparison with analytical solutions, it will be neglected in the following Sections II, III and V. However, it will be considered in the section VI.

II. FREE-CARRIER DENSITY: VALIDATION OF ANALYTICAL CASES

A. Auger recombination

The formulation proposed by [?] has to be validated within several steps. Auger recombination is modeled by a coupled non-linear system. In that section, we only consider the equation of free-carrier given by

$$\frac{\partial n_e}{\partial t} = \frac{\partial n_h}{\partial t} = C_h^{AR} n_h^2 n_e + C_e^{AR} n_e^2 n_h \quad (5)$$

Analytical solution of the equation is given by

$$n_e(t) = \frac{C_{AR}^{(h)} n_e^0 n_h^0}{\left(n_e^0 C_{AR}^{(e)} + n_h^0 C_{AR}^{(h)} \right) \exp \left(C_{AR}^{(h)} n_h^2 t \right) - C_{AR}^{(e)} n_e^0}$$

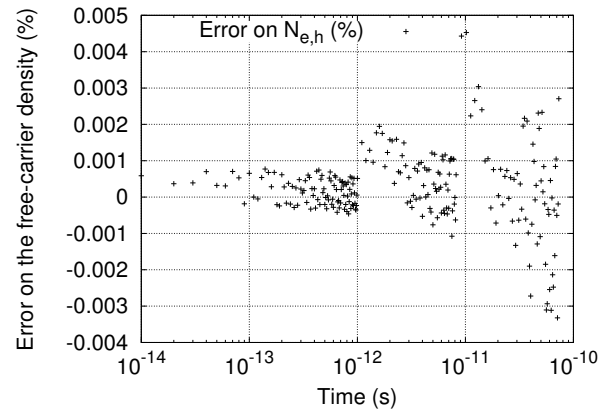


Figure 1. Relative error calculation (%) of Eq. (5) as a function of time.

Figure 1 presents the comparison between the numerical calculation and the analytical solution for an initial value problem, where densities have been fixed at $N_{e,h}(t=0) = 10^{27} \text{ m}^{-3}$. The mismatch between numerical and analytical calculations is smaller than 0.005%.

B. Interband absorption

d. One-photon absorption Validation on analytical function at $\lambda = 343 \text{ nm}$ has been performed but is not yet presented.

e. Two-photon absorption Test at $\lambda = 1030 \text{ nm}$. Test is shitty ! Go to second order time integration scheme.

C. Impact ionization

To be completed.

III. FREE-CARRIER ENERGY: VALIDATION OF ANALYTICAL CASES

A. Heating of the interband absorption

In this subsection, we simply consider the interband absorption at 343 nm.

$$\frac{\partial}{\partial t} (C_e T_e) = \frac{\hbar\omega - E_{gap}}{\hbar\omega} \sigma_1 I$$

where the free-carrier density has been fixed to $n_e = 10^{27} \text{ m}^{-3}$.

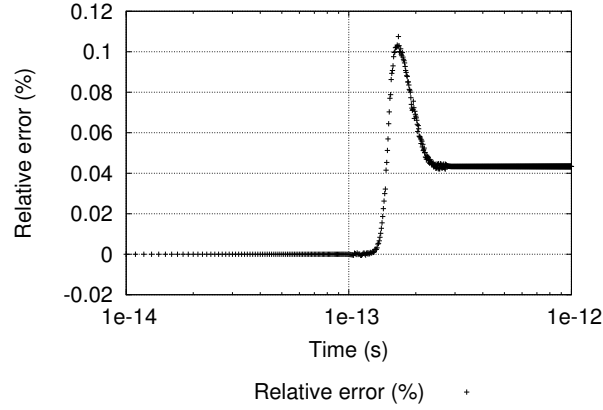


Figure 2. Relative error between numerical and analytical solutions for the interband absorption of electrons. $\lambda = 343 \text{ nm}$.

Figure 2 shows the relative error between the numerical and the analytical solution. The maximum relative error is 0.1%, which is reasonable. This relative error can be reduced by replacing the first-order upwind time integrator by a second-order one.

At 1030 nm, the two-photon absorption plays an important role. It is thus of interest to validate the integrator at this wavelength. For that purpose, we push the laser fluence to 100 J.m^{-2} in order to reach a good 2-photon absorption probability. Figure 3 presents the free-carrier heating by the 2-photon absorption process at $n_e = 10^{27} \text{ m}^{-3}$.

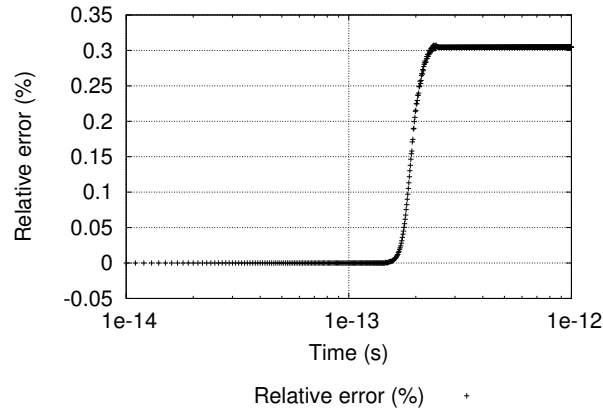


Figure 3. Relative error between numerical and analytical solutions for the calculation of 2-photon absorption at $\lambda = 1030 \text{ nm}$ wavelength, $F = 100 \text{ J.m}^{-2}$ and pulse duration $\tau = 40 \text{ fs}$.

B. Validation of the electron energy coupling with lattice

f. Cooling of the electrons by the cold lattice In this section, the equations are reduced to the following

$$\frac{\partial}{\partial t} (C_e T_e) = -\gamma_{e-ph} (T_e - T_s) \quad (6)$$

$$\gamma_{e,h-ph} = C_{e,h} \nu_{e,h-ph}. \quad (7)$$

$$\nu_{e,h-ph}^{-1} = \tau_{ph}^0 \left[1 + \left(\frac{n_{e,h}}{n_{th}} \right)^2 \right]$$

Free-carrier density n_e is taken as a constant $n_e = 10^{27} \text{ m}^{-3}$. The initial carrier temperature is set to the value of $T_e = 300,000 \text{ K}$.

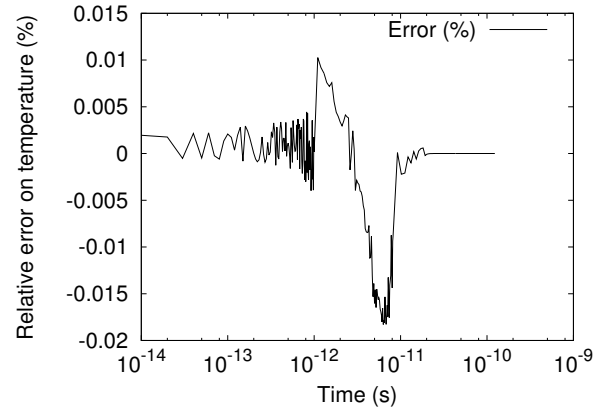


Figure 4. Relative error on the calculation of the free-carrier temperature. In this result, only the thermal coupling between electrons and lattice has been checked. The sudden change of the relative error is due to the change of timestep during the calculation.

Figure 4 presents the relative error between the numerical and the analytical solution as a function of time. Results show that the free-carrier temperature is calculated with an error lower than 0.01 % from the exact value. The effect of the change of time-step is also observed: the change at $t = 10^{-12} \text{ s}$ induces a less precise calculation, which is then amplified until the end of the interaction. The overall error remains however lower than 0.015 %, which is rather good.

IV. VALIDATION OF LATTICE THERMAL COUPLING

A. Heating of the lattice by hot electrons

g. Reduction to a “simple” case This section investigates the heating of the lattice in our conditions. Eq. 3 is reduced to its simplest form allowing to validate to calculate the heating of the lattice. The laser source is disabled, and we only consider the coupling between a gas of hot electrons at a fixed density coupled with the lattice. This allows to set the carrier density n_e as a constant, and the free-carrier temperature T_e as a constant. Thus, the carrier heat capacity C_e , and the electron-lattice coupling rate γ_{e-ph} are constant. These approximations lead to set the following equation

$$\frac{\partial [C_s(T_s(t)) \times T_s(t)]}{\partial t} = \gamma_{e-ph} [T_e^0 - T_s(t)] \quad (8)$$

where

$$\gamma_{e,h-ph} = C_{e,h} \nu_{e,h-ph}. \quad (9)$$

$$\nu_{e,ph}^{-1} = \tau_{ph}^0 \left[1 + \left(\frac{n_e^0}{n_{th}} \right)^2 \right].$$

In our case, the lattice heat capacity strongly depends to the lattice temperature T_s . Depending on the form of the coefficient $C_s(T_s)$, Eq. 8 will therefore exhibit strong non-linearities. Before considering solving this equation, we first develop the possible functions used to adjust the lattice heat capacity on the available experimental results. In all the cases, the equation is non-linear and does not admit a simple analytical solution. We are thus able to compare the quality of the numerical integration with other numerical integration packages and keep checking the energy conservation of our results. However, no error estimation will be possible to perform since no analytical solution is available.

h. Adjustment of the lattice heat capacity Here is a list of the functions which have been built as a function of lattice temperature fitting the experimental measurements of Refs. [1, 2]. Each case will lead to different types of non-linearities. Since we have different possibilities to adjust the experimental data, the type of function should be chosen in order to minimize the difficulties over the numerical integration.

1. Exponent fraction

$$C_s(T_s) = \frac{a_0}{e^{a_1/T_s} - a_2} = \frac{0.4129 \times 10^3 \rho_{Si}}{e^{\frac{88.183}{T_s}} - 0.6765}$$

The resulting type of the equation is

$$a_0 a_1 \left(\frac{\partial T_s}{\partial t} \right) e^{\frac{a_1}{T_s(t)}} \left(e^{\frac{a_1}{T_s(t)}} - a_2 \right)^{-2} (T_s(t))^{-1} + a_0 \frac{\partial T_s}{\partial t} \left(e^{\frac{a_1}{T_s(t)}} - a_2 \right)^{-1} = \frac{C_e (T_e^0 - T_s(t))}{\tau_e}$$

2. Sum of exponents

$$C_s(T_s) = \eta (ae^{bT_s} + ce^{dT_s})$$

where $\eta = 10^3 \rho_{Si}$, $a = 0.899$, $b = 5.455 \times 10^{-5}$, $c = -0.959$, and $d = -0.004218$. The resulting type of equation is

$$\left(\frac{\partial T_s}{\partial t} \right) \eta \left(T_s(t) a b e^{b \times T_s(t)} + T_s(t) c d e^{d \times T_s(t)} + a e^{b \times T_s(t)} + c e^{d \times T_s(t)} \right) = \frac{C_e (-T_e^0 + T_s(t))}{\tau_e}$$

3. Real-valued polynomial

$$C_s(T_s) = \sum_{i=0}^{i=5} a_i T_s^i$$

where $a_5 = 2.36 \times 10^{-16}$, $a_4 = -1.707 \times 10^{-12}$, $a_3 = 4.619 \times 10^{-9}$, $a_2 = -5.912 \times 10^{-6}$, $a_1 = 3.733 \times 10^{-3}$, and $a_0 = -0.0494$. The resulting of equation is

$$\left(\frac{\partial T_s}{\partial t} \right) \left(2a_1 T_s(t) + 3a_2 (T_s(t))^2 + 4a_3 (T_s(t))^3 + 5a_4 (T_s(t))^4 + 6a_5 (T_s(t))^5 + a_0 \right) = \frac{C_e (-T_e^0 + T_s(t))}{\tau_e}$$

4. Van Driel's style fit function

$$C_s(T_s) = 10^3 \rho_{Si} \times (a_0 T_s + a_1 T_s^{a_2})$$

with $a_0 = -0.003592$, $a_1 = 0.01458$, and $a_2 = 0.8316$. The stability of this function is very good. However, the temperature validity range of this function is limited to 1400 K.

i. Numerical integration Several possibilities can be tried to deal with the non-linearity.

1. Use a higher-order precision scheme on the time integration for the T_s equation.
2. Use a Runge-Kutta solver. However, there gonna be diffusive terms enabled later, which will not be mathematically clean with such a scheme.
3. Iterate the solution and ensure convergence *before going to the next time-step* (can really multiply the calculation time).

Applying a higher-order precision scheme allows to obtain solutions closer to the “exact” solution. Here, the “exact” solution is considered to be the one given by the 4th order Runge Kutta scheme (“ode45” function of Maple and Matlab). However, using an eulerian scheme does not provide a sufficiently good convergence of the solution even at the 4th order. The choice of the fitting function using the Van Driel's form interestingly allows to obtain a solution equal to the “exact” one. The origin of this numerical recipe is not understood yet.

It should be mentioned that in the all the cases considered in this section, the energy conservation has been verified and did not show any deficiency. The serious non-linearity that we face in this section is important for the topic and can strongly affect the final results given by the code.

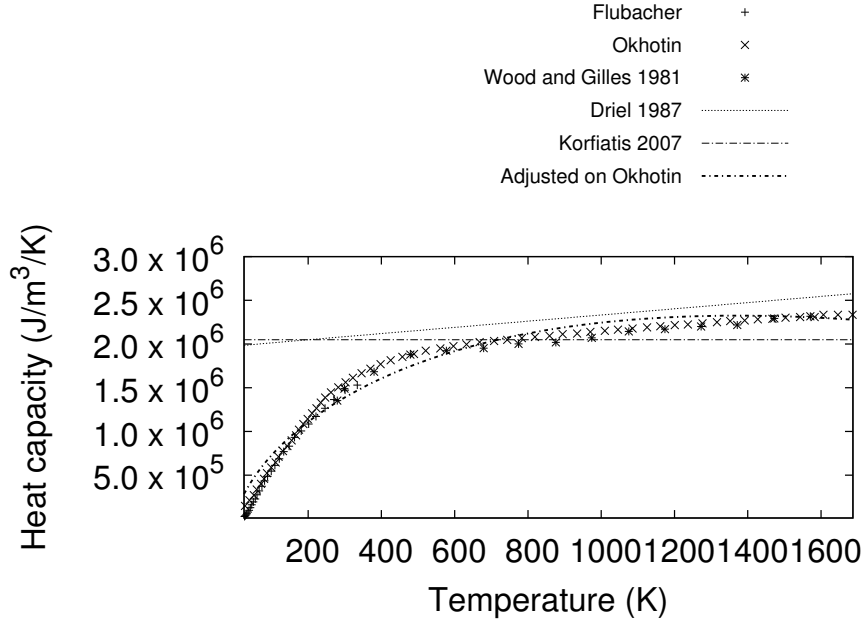


Figure 5. Heat capacity of Si and various adjustments.

V. VERIFICATION OF THE COUPLED EFFECTS

A. Change of carrier heat capacity

B. Auger recombination

$$\frac{\partial n_e}{\partial t} = \frac{\partial n_h}{\partial t} = C_h^{AR} n_h^2 n_e + C_e^{AR} n_e^2 n_h$$

$$\frac{\partial}{\partial t} (C_e T_e) = E_{gap} C_{AR}^{(e)} N_e^2 N_h \quad (10)$$

$$\frac{\partial}{\partial t} (C_h T_h) = E_{gap} C_{AR}^{(h)} N_h^2 N_e \quad (11)$$

If $n_e(t=0) = n_h(t=0)$, then $n_e(t) = n_h(t)$. The system is thus equivalent to the following

$$\begin{cases} \frac{\partial n_e}{\partial t} &= C_{AR}^{(e)} n_e^3 \\ \frac{\partial}{\partial t} (C_e T_e) &= E_{gap} C_{AR}^{(e)} N_e^3 \end{cases} \quad (12)$$

with the initial conditions $T_{e,h}(t=0) = T_{e,h}^{(0)}$ and $N_e(t=0) = N_e^{(0)}$. The solution of the Eq. (12) is

$$\begin{cases} N_e(t) &= \left(\frac{1}{N_{e,0}^2} + 2C_{AR}^{(e)} t + 2C_{AR}^{(h)} t \right)^{-\frac{1}{2}} \\ T_e(t) &= \frac{\left\{ \sqrt{2t \left(C_{AR}^{(e)} + C_{AR}^{(h)} \right) + N_{e,0}^{-2} \left[3k_B T_e N_{e,0} \left(C_{AR}^{(e)} + C_{AR}^{(h)} \right) + 2E_g C_{AR}^{(e)} N_{e,0} \right] - 2E_g C_{AR}^{(e)}} \right\}}{3k_B \left(C_{AR}^{(e)} + C_{AR}^{(h)} \right)} \end{cases}$$

In that calculation, the band gap energy is taken constant.

Figure 6 shows the relative error ζ between the numerical and analytical results, calculated for $T_{e,h}$ and $N_{e,h}$ using the following formula

$$\zeta(T_e) = 100 \times \frac{T_e^{(num)} - T_e^{(exact)}}{T_e^{(exact)}}$$

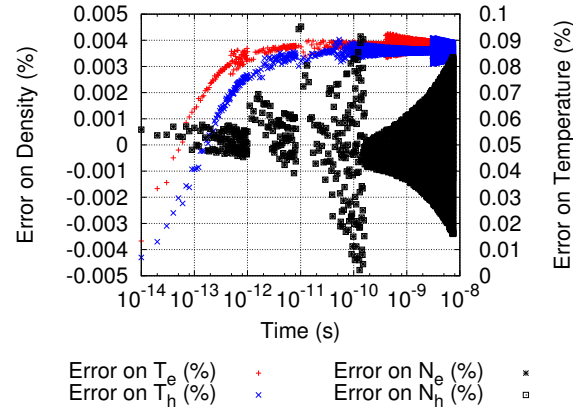


Figure 6. Calculation of the error (%) made on the solutions $N_{e,h}(t)$ and $T_{e,h}(t)$ for $N_{e,h}(t=0) = 10^{27} \text{ m}^{-3}$, $T_{e,h}(t=0) = 80 \text{ K}$.

The first conclusion is that the Auger recombination is very well solved. Calculating on a time range of 10^{-8} s with a timestep of $dt \sim 10^{-17} \text{ s}$, the relative error is lower than 0.1 % on the carrier temperature, and lower than $5 \times 10^{-3} \%$ on the carrier density. The multiple time scale of the problem is taken into account by changing the timestep during the calculations. In this calculation, the timestep has been changed at $t = 10^{-12} \text{ s}$, $t = 10^{-11} \text{ s}$, and $t = 10^{-10} \text{ s}$ by multiplying $dt' \leftarrow 10 \times dt$. We can directly observe the negligible consequences of the timestep modification as a function of time.

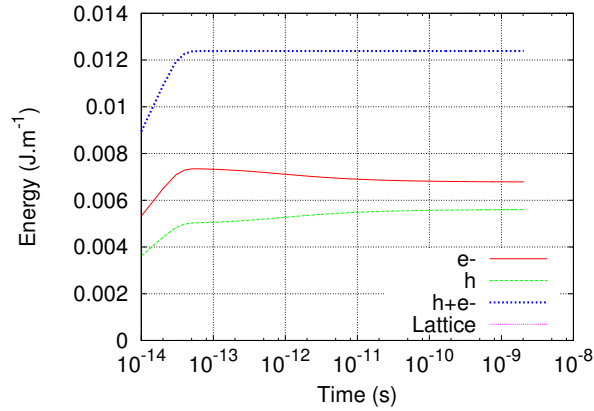


Figure 7. Energy of the free-carriers as a function of time after the maximum of the laser pulse. $F = 10 \text{ J.m}^{-2}$, $\tau = 40 \text{ fs}$, $\lambda = 343 \text{ nm}$. It is worth to note that considering Fermi-Dirac statistics leads to the same result.

Figure 7 presents the conservation of the carrier energy after their excitation by 1 photon absorption. The Fermi-Dirac statistics are here taken into account. It is shown that the free-carrier energy is well conserved.

VI. CONTROL OF THE ENERGY CONSERVATION

Since the mesh is not regular and that finding analytical solution to the equation using an arbitrary contour in a reasonable duration is beyond the capabilities of the author, it has been decided to study the energy conservation as a function of time. The timescale of the processes investigated by this code is near from the nanosecond reaction of the system, which is 6 orders of magnitude larger than the laser pulse duration (of about 40 fs to 500 fs). The validity of the time-integrator is thus of major importance in this multi-scale problem, the connexion to the spatial steps is thus straightforward.

A. Energy of the carriers

Since we model an external laser pulse irradiating the sample by its intensity evolution with time and space through the function $I(t, x, y)$, then we will express the total energy considering the invariance by translation with the z axis, normal to the orientation of this document plane.

The total energy study will be expressed in J/m . Actually, the energy of the carriers in 2D is defined by

$$dE_e [J.m^{-3}] = dE_{e,kin} + dE_{pot} = d(C_e T_e) + d(E_g n_e)$$

where $\mathcal{V}$ [m^2] is the control volume of each 2D cell of the mesh, $E_{e,kin}$ expresses the thermal energy of conduction band electrons, $E_{e,pot}$ expresses the potential energy of the conduction band electrons. This latter expression can be also written for holes. However, since the origin of the potential energies is taken at the edge of the valence band, then holes have a potential energy equal to 0. Thus we can write

$$dE_h [J.m^{-3}] = dE_{h,kin} + dE_{h,pot} = d(C_h T_h)$$

Integrating over the two dimensions of the problem leads to write the total energy $\xi_{e,h}$ at a given instant in the whole sample volume by

$$\begin{aligned} \xi_e [J.m^{-1}] &= \iint dE_e d\mathcal{V} \\ &= \iint (T_e dC_e + C_e dT_e + dE_g n_e + E_g dn_e) d\mathcal{V} \end{aligned}$$

$$\begin{aligned} \xi_h [J.m^{-1}] &= \iint dE_h d\mathcal{V} \\ &= \iint (T_h dC_h + C_h dT_h) d\mathcal{V} \end{aligned}$$

Band gap is a function of lattice temperature T_s and carrier density n_e expressed via [3]

$$E_g(T_s, n_e) = 1.17 - 4.73 \times 10^{-4} \frac{T_s^2}{T_s + 636} - 1.5 \times 10^{-10} n_e^{1/3}$$

B. Control of the laser energy

Now we have chosen the unity to express the energy in the problem, we express now the total laser energy is by the expression

$$\xi_{sample} [J.m^{-1}](y) = \iint I(t, x, y) dt dx$$

integrated over the diameter of the laser spot.

However, the shape of the surface is not flat, and the surface reflectivity is taken into account by the analytical solutions of Mie scattering, assuming a scattering on a 3D cylinder from which a plane section is studied: the intersection between the study plane and the entire 3D space. Consequently, the calculation of the overall reflectivity and transmittance by the problem shape is rather difficult.

1. Calculation of the tip absorptance

We try to solve this problem by studying the energy absorbed by the nanometric sized sample via

$$\xi_{absorbed} [J.m^{-1}] = \iiint \alpha I(t, x, y) dt dx dy$$

In the conditions of the experiments ($\tau = 40$ fs, $F = 10$ J/m², $\lambda = 343$ nm), the order of magnitude of $\xi_{absorbed}$ is 10^{-4} J. In order to calculate the overall absorbed laser energy, we define

$$A = \frac{\xi_{absorbed}}{\xi_{laser}}$$

where

$$\xi_{laser} = F \times \mathcal{V}$$

M	N	MeshShift	$\mathcal{V}$	dt	$\xi_{absorbed}$ (J)	$A(\%)$
11	11	1	6.7004×10^{-12}	10 as	5.2097×10^{-11}	77.75
11	11	4	6.72711×10^{-12}	"	5.5004×10^{-11}	81.76
51	51	4	6.73268×10^{-12}	"	6.1999×10^{-11}	92.08
101	101	4	6.73404×10^{-12}	"	6.4153×10^{-11}	95.26
101	101	10	6.73415×10^{-12}	"	6.4271×10^{-11}	95.44
101	101	20	6.73432×10^{-12}	"	6.4122×10^{-11}	95.21
151	151	4	6.73439×10^{-12}	"	6.4955×10^{-11}	96.45
151	151	20	6.7345×10^{-12}	"	6.4528×10^{-11}	95.81

Table I. Optical energy as a function of the mesh precision for an homogeneous energy distribution $\nabla I(x, y) = 0$, where $F = 10$ J.m⁻², and $\tau = 40$ fs.

Table I presents the total optical energy deposited in the sample as a function of the mesh properties in case of an homogeneous distribution of laser energy. This allows to calculate the minimal mesh and timestep parameters required to reach convergence in the numerical sample volume and the convergence of the intensity calculation.

2. Calculation of the incident laser energy

The intensity at the tip apex interface is defined by the following formula

$$E_{laser} [J] = \iint \vec{S} \cdot \vec{n} d\mathcal{A} dt$$

where $\vec{S}$ is the Poynting vector, $\vec{n}$ is the normal to the 3D surface of the cone, and $\mathcal{A}$ is the area element defined around the normal vector. The intensity is defined by the flux of the Poynting vector by the surface of area $d\mathcal{A}$. In the plane of the simulation (xOy), we can write the following relation

$$\vec{S} \cdot \vec{n} = I [x, y_{\partial\Omega}] \cos \theta_{\partial\Omega}$$

where

$$\tan \theta_{\partial\Omega} = \frac{\partial x}{\partial y}.$$

Since we consider only two dimensions with an invariance by translation along the coordinate z , we can build the quantity

$$\xi_{laser} [J.m^{-1}] = \iint \vec{S} \cdot \vec{n} d\mathcal{L} dt$$

using the contour of the cone in the plane of the model (xOy), where the projection of the Poynting vector at the interface is known.

The arc length along the curve $x(y)$ is given by

$$\mathcal{L} = \int \left| \frac{\partial x}{\partial y} \right| dy$$

Therefore, we can write

$$\xi_{laser} [J.m^{-1}] = \iint I [x_{\partial\Omega}(y), y, t] \cos \left(\arctan \frac{\partial x}{\partial y} \right) \left| \frac{\partial x}{\partial y} \right| dy dt.$$

For a constant intensity defined by $I(x, y, t) = I_0 e^{-\frac{1}{2}(\frac{t}{\sigma_t})^2}$, $\sigma_t = \frac{\tau}{2\sqrt{2\ln 2}}$, $I_0 = \frac{F}{\tau} \sqrt{\frac{4\ln 2}{\pi}}$, $F = 10 J.m^{-2}$, and $\tau = 40$ fs, it is found

3. Verification of the calculated intensity

$$\xi_{laser} = \iint (\sigma_1 + \alpha_{fcr,e} + \alpha_{fcr,h}) I_0(t, x, y) d\Omega dt$$

$$\xi_{opt} = \iint (\sigma_1 + \alpha_{fcr,e} + \alpha_{fcr,h}) I(t, x, y) d\Omega dt$$

We define the numerical trust value which corresponds to the ratio between the energy absorbed contained the solid, with the energy given by the laser intensity distribution.

$$\eta = \frac{\xi_{solid}}{\xi_{opt}}$$

We also define the absorptivity by

$$\overline{A} = \frac{\xi_{opt}}{\xi_{laser}}$$

which is the effective absorptivity of the laser pulse considering the shape of the cone.

a. *Results* Table II shows the results for 343 nm using different meshes. It is clear that :

- The “numerical trust index” η demonstrates that the size of the mesh is of major importance for obtaining a good energy conservation between the field and the solid. But why, since the equations are using the same mesh ? It can be due to the fact that the strongest absorption zone at 343 nm is at the boundary, and that the first energy contained in the cells of the boundary is neglected. Thus, at 515 nm, we shall need less cells to reach the energy balance between the fields and the solid.
- the effect of the meshgrid is negligible in the energy balance. It will take its importance only for the energy diffusion conservation.
- A problem is identified in the definition of the optical absorption which simply cannot be so small. Indeed, at $\lambda = 343$ nm, the penetration depth of the field is nearly 9 nm, which is clearly smaller than the size of the cone. Thus, it is not possible that the absorbed energy is so small. Its definition might be wrong or reveals a numerical problem.

VII. GENERATION AND CONTROL OF THE MESH

b. *Definition of the contour*

c. *Regularization of the mesh* The mesh is built by defining a contour and solving

$$\Delta \begin{pmatrix} x \\ y \end{pmatrix} = 0$$

by a simple iterative method

$$x_i = \frac{1}{4} (x_{i-1,j-1} + x_{i-1,j+1} + x_{i+1,j-1} + x_{i+1,j+1})$$

M	N	MeshShift	$\mathcal{V}$	dt	λ (nm)	Pola	ϕ (rad)	ξ_{laser} (mJ.m ⁻¹)	ξ_{opt} (mJ.m ⁻¹)	ξ_{solid} (mJ.m ⁻¹)	Trust η (%)	$\bar{A} = 1 - R - T$ (%)
101	101	4	6.73404×10^{-12}	"	343	TM	0	7.384	33.71×10^{-3}	13.75×10^{-3}	40.7	0.46
101	101	10	6.73415×10^{-12}	"								
101	101	20	6.73432×10^{-12}	"								
151	151	4	6.73439×10^{-12}	"	343	TM	0	7.384	29.53×10^{-3}	18.07×10^{-3}	61.2	0.40
251	251	4	6.73462×10^{-12}	"	343	TM	0	7.384	27.093×10^{-3}	21.808×10^{-3}	80.49	0.37
351	351	40	6.73472×10^{-12}	"	343	TM	0	7.384	26.340×10^{-3}	23.061×10^{-3}	88.95 (no diff.)	0.356

Table II. Optical energy as a function of the mesh precision for an inhomogeneous energy distribution $\nabla I(x, y) \neq 0$, where $F = 10 \text{ J.m}^{-2}$, and $\tau = 40 \text{ fs}$. “Trust” is defined by $\eta = \frac{\xi_{solid}}{\xi_{opt}}$.

$$y_i = \frac{1}{4} (y_{i-1,j-1} + y_{i-1,j+1} + y_{i+1,j-1} + y_{i+1,j+1})$$

Round about 70 000 iterations are required to reach convergence of the mesh.

This method is optimum on a single core since convergence rate of this technique is rather good (how to quantify ?) in respect of the global calculation time.

VIII. OPTIMIZATION OF THE CODE

Optimization is a big word for telling many things. First, we have designed a model which is actually already simplified to decrease the amount of necessary operations. In this section, purpose is to perform optimizations over the code, especially over the way to perform the operations on a single processor, and to decrease the waste of operations.

A. Sequential code: checking the bottlenecks

Using the Intel Loop Observer software (or GNU profiler), we can monitor the number of calls to the loops and number of calls to the functions.

By performing a first test during one night, I already figured out that a function was using 25% of the calculation time (sequential) despite that it was not necessary. The number of operations was found to be 10^{12} after one night. Actually, this operation was required once, and then replaced by calls to memory. But is this call to memory really saves time instead of performing such simple operation each time ?

In the next section, we focus on optimizing the way to perform calculations with several cores, to parallelize the operations. First, we will define the tools used to quantify the efficiency of a parallel code, and we will also perform some verification allowing to avoid the unnecessary operations.

B. Calculation of the code velocity

$$CPU = \frac{N_{timesteps}}{t_{current} - t_{initial}} \times N_{threads}$$

where $t_{current}$ is the CPU time captured just before performing the outputs, and $t_{initial}$ is the CPU time captured at the initialization of the code. Hence, CPU is supposed to be constant within the execution time of the code.

The coefficient CPU is built to be independent to number of cores used. In this purpose, it becomes possible to calculate the speedup η in core-number-independent way. Of course, the speedup may increase while increasing the number of cores, hopefully, until saturation.

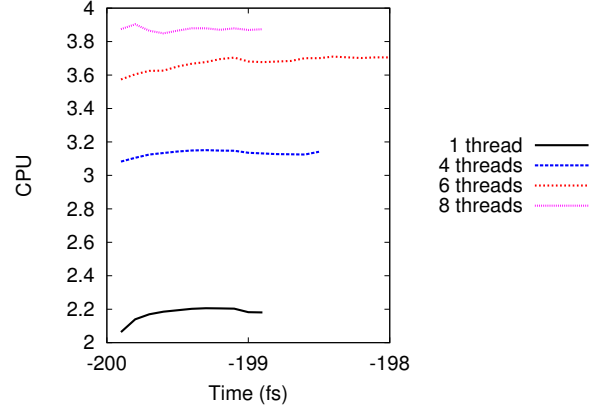


Figure 8. CPU efficiency as a function of core number. Note that these calculations were performed with a homogeneous energy source in the sample.

One understands that the definition of CPU is dependent to the frequency of the CPU clock. Hence, it is not reasonable to compare the CPU quantity between different computers. However, we can normalize the CPU by the velocity of the code in sequential mode on a single computer in order to build the speed up η , a quantity that it is intrinsic to the efficiency of the code.

$$\eta(N) = \frac{CPU_{N-cores}}{CPU_{1-core}}$$

Hence, it looks that speedup of our parallelization method is greater than 1, but is lower than 2 with the available number of cores.

C. Changing the order of the loops

The main loop of the code is consuming 30% of the calculation time in sequential mode. By changing the order of the loops, and by parallelizing the inner or outer ones, we can obtain serious improvement over the speedup. Fig. 9 demonstrates a very strong influence over the order of the loop iterations and distribution. Despite that one solution looks to be more efficient than the others, it is not sure if such optimization conserves the physical meaning of the result.

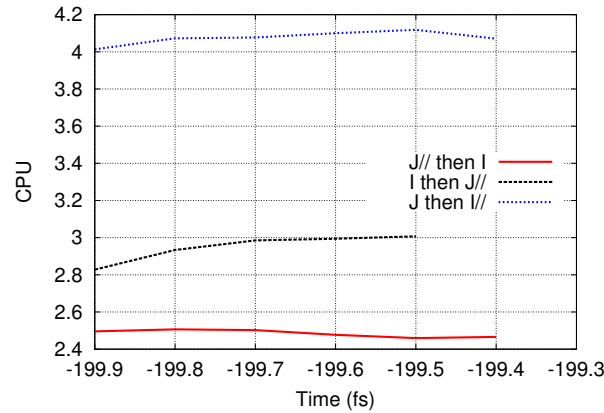


Figure 9. Qualitative increase of the efficiency as a function of the main loop order in Flaps2D.

D. Parallel code: reduce the number of conditional branches

IF instruction is consuming a lot of time. Hence, it must be removed, especially from loops. Problem is that a large code allows to perform many types of calculations and to select various parameters. How can we select sub-modes of calculations into a code without using condition branches ?

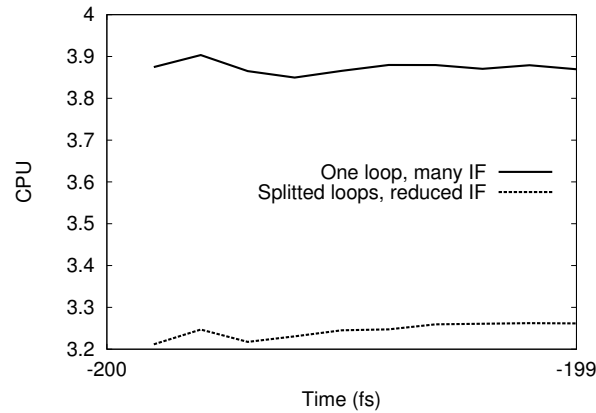


Figure 10. Comparison before and after reduction of the IF branches inside parallelized loops.

Figure 10 indicates that actually, cutting the loops into one parallel loop per equation is not the best idea. Indeed, the physical quantities are independent, since we use an explicit solving scheme. Hence, increasing the number of parallel loops increase the number of synchronization delays expected at the end of solving each equation. It is then more efficient to parallelize one mesh node per loop and to solve everything in this mesh node using the same core.

E. Efficiency of the parallelization method

d. Distribution of the loops on CPU cores The method of distributing the loops over the various cores has a limited efficiency. However, we can try to optimize the way they are distributed with the command `OMP_SCHEDULE`.

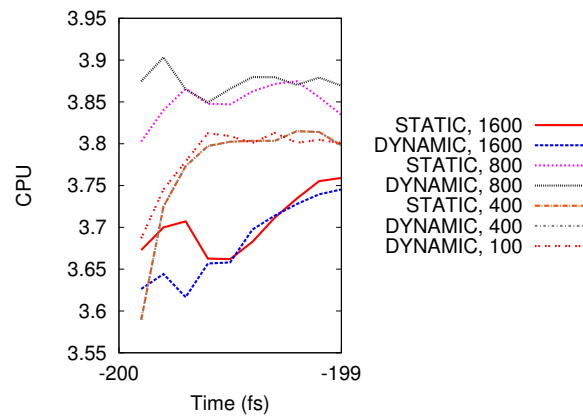


Figure 11. CPU efficiency as a function of the method to distribute the loops with OpenMP using 8 cores.

Figure 11 presents the CPU efficiency evolution as a function of the strategy to distribute the loops over the cores. These calculations were performed only using 8 cores.

e. Explicit integration VS implicit integration Overall the optimization performed until now, Figure 12 presents the efficiency of the code for the implicit version and the explicit version, including recent optimizations. The implicit version of the code consumes round about 7 GB of RAM, and the explicit version consumes ~ 400 MB.

f. Which parallelization method can be investigated to strongly increase the speedup ?

1. Go to GPU.

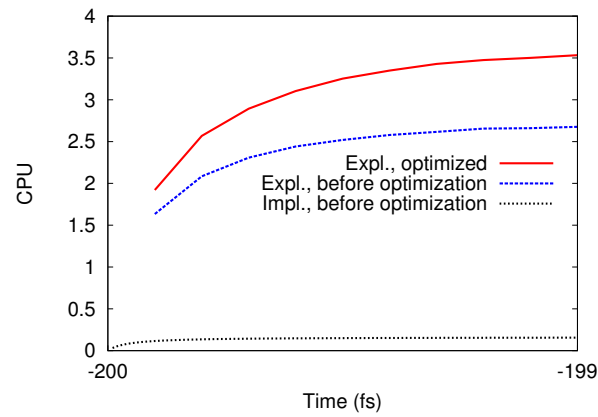


Figure 12. CPU efficiency as a function of the method to distribute the loops with OpenMP using 8 cores.

2. Find OpenMP bottlenecks using OpenMP profiler program.
3. Try the code on 16 cores CPU to verify scalability.

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